

Three-phase C&I Hybrid Inverter



X3-ULTRA

15kW / 19.9kW / 20kW
25kW / 30kW



Smart Management

- Single unit UPS-level switchover time <10ms
- Built-in shadow tracking
- Smart loads management(e.g. heat pump, smart EV charger)
- Loads respond time within 0.3 s
- VPP ready with a variety of compatibility (OpenADR, IEEE2030.5, FCAS, API)*



High Performance

- 200% PV oversizing and up to 110% AC output
- 200% EPS overload for 10s
- Max. 60A charging / discharging current
- Low start-up voltage for more power generation



Assured Reliability

- IP66 Ingress protection
- Type II SPD on AC&DC side
- Optional AFCI protection

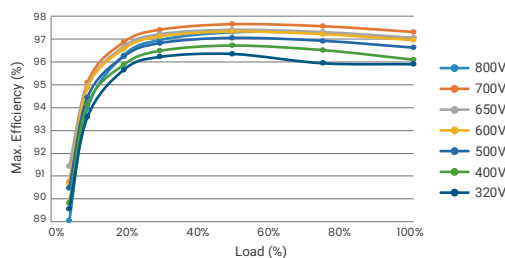


Flexible Adaptability

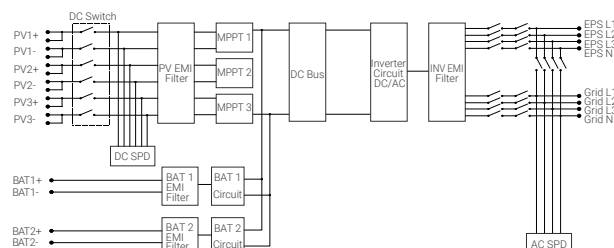
- Max. 10pcs parallel for on-grid and off-grid
- Microgrid and generator function for versatile operations
- Max. 36A PV input per MPPT, optimized for high-power solar panel

* Feature to be upgraded in the future

Efficiency Curve



Circuit Diagram



X3-ULT-15K X3-ULT-15KP X3-ULT-19.9K X3-ULT-20K X3-ULT-20KP X3-ULT-25K X3-ULT-30K

PV INPUT						
Max. recommended PV array power	30 kWp		40 kWp		50 kWp	60 kWp
Max. PV input voltage ^①	1000 V					
Rated PV input voltage	600 V					
Operation voltage range	120 ~ 950 V					
MPPT voltage range ^②	160 ~ 950 V					
Start-up voltage	200 V					
No. of MPP trackers / strings per MPP tracker	2 / (2 / 2)	3 / (2 / 2 / 2)	2 / (2 / 2)		3 / (2 / 2 / 2)	
Max. input current per MPPT	36 A / 36 A	36 A / 36 A / 36 A	36 A / 36 A		36 A / 36 A / 36 A	
Max. input short circuit current per MPPT	45 A / 45 A	45 A / 45 A / 45 A	45 A / 45 A		45 A / 45 A / 45 A	
AC INPUT & OUTPUT (ON-GRID)						
Rated output power	15000 W (AS4777 14999 W)		19999 W	20000 W	20000 W	25000 W (VDE4105 24900 W) 30000 W (AS4777 29999 W, VDE4105 29900 W)
Rated output current	21.8 A		29.0 A	29.0 A	29.0 A	36.3. A 43.5 A
Max. output apparent power	16500 VA (AS4777 14999 VA)		19999 VA	22000 VA	22000 VA	27500 VA (VDE4105 24900 VA) 30000 VA (AS4777 29999 VA, VDE4105 29900 VA)
Max. output continuous current	24.0 A (AS4777 21.8 A)		29.0 A	31.9 A	31.9 A	39.9 A (VDE4105 36.3 A) 43.5 A
Rated AC voltage	3 / N / PE, 220 / 380 V 3 / N / PE, 230 / 400 V					
Max. AC input apparent power	15000 VA		19999 VA	20000 VA	20000 VA	25000 VA 30000 VA
Max. AC input current	21.8 A		29.0 A	29.0 A	29.0 A	36.3 A 43.5 A
Rated AC frequency	50 Hz / 60 Hz					
Adjustable power factor range	~ 1 (0.8 lagging to 0.8 leading)					
THDi (rated power)	< 3%					
BATTERY						
Battery type	Lithium					
Battery voltage range ^{③④}	120 ~ 800 V					
Max. charge / discharge current	60 A (30 A × 2)					
EPS (OFF-GRID) OUTPUT (WITH BATTERY)						
Rated EPS output voltage, frequency	230 V / 400 V, 50 Hz / 60 Hz					
Rated EPS output power	15000VA		19999 VA	20000 VA		25000 VA 30000 VA
Peak EPS output power	2 times of rated power, 10 s					
Switchover time	< 10 ms					
EFFICIENCY						
Max. efficiency	98.0%					
European efficiency	97.7%					
ENVIRONMENT LIMIT						
Ingress protection	IP66					
Operation temperature range	-35 ~ 60°C (> 45°C derating)					
Max. operation altitude	3000 m					
Relative humidity	0 ~ 100% RH (condensing)					
Overvoltage category	Mains: III, Battery: II, PV: II					
GENERAL						
Dimensions (W × H × D)	696 × 526 × 240 mm					
Net weight	47 kg					
Cooling concept	Smart air cooling					
Communication interfaces	Meter (RS-485), DI x 2, DO x 1, Modbus					
Power consumption (night)	< 5 W					
Topology	Non-isolated					
Certifications	VDE4105, G99, AS4777, EN50549, CEI 0-21, IEC61727, PEA/MEA, NRS-097-2-1, RD1699, TOR					
PROTECTION						
Protections	Over / under voltage protection, DC reverse-polarity protection, Residual current detection, Over temperature protection, DC isolation protection, Grid monitoring, DC injection monitoring, Back feed current monitoring					
Active anti-islanding method	Frequency shift					
Surge protection	DC: Type II, AC: Type II					
Arc-fault circuit interrupter (AFCI)	Optional					
Protective Level	Class I					

① The maximum input voltage is the upper limit of the DC voltage. Any higher input DC voltage would probably damage inverter

② Input voltage exceeding the MPPT voltage range may triggers inverter protection

③ Compatible with a minimum of 3 units of HS Series batteries, but if the total voltage of the 3 batteries is less than 127V and there is no PV input, the system will not able to startup

④ When the voltage is below 180V, the inverter will limit the battery current to less than 20A